BCA Lab Practical File (Oops Using Java)

Unit I – Java Basics and Programming

1. Write a Java program to print "Hello World".
2. Write a Java program to display your name, roll number, and course.
3. Write a Java program to add, subtract, multiply, and divide two numbers.
4. Write a Java program to swap two numbers using a third variable.
5. Write a Java program to calculate the area of a circle.
6. Write a Java program to check whether a number is even or odd.
7. Write a Java program to find the largest of three numbers.

Unit II – Classes, Objects, Methods and Inheritance

1. Create a Student class with data members name, roll number, and marks. Display student details using an object.
2. Write a Java program to demonstrate a parameterized constructor.
3. Write a Java program to demonstrate method overloading.
4. Write a Java program to demonstrate static variables and static methods.
5. Create a class Animal and inherit it into a class Dog. Display their properties using inheritance.
6. Create an abstract class Shape with an abstract method area(). Implement it in Circle and Rectangle classes.

Unit III – Interfaces, Packages, Arrays and Strings

1. Write a Java program to demonstrate extending an interface.
2. Write a Java program to create a package named `mypackage` and access its class from another program.
3. Write a Java program to create and display a one-dimensional integer array.
4. Write a Java program to find the largest and smallest elements in an array.
5. Write a Java program to calculate the sum and average of array elements.

Unit IV – Exception Handling and Multithreading

1. Write a Java program to demonstrate exception handling using try and catch.
2. Write a Java program to handle ArithmeticException when dividing a number by zero.
3. Write a Java program to demonstrate multiple catch blocks.
4. Write a Java program to create and throw your own exception using throw.
5. Write a Java program to demonstrate the use of the throws keyword.
6. Write a Java program to create a thread by extending the Thread class.

Unit V – File Handling and Input/Output

1. Write a Java program to create a new file using the File class.
2. Write a Java program to check whether a file exists or not.
3. Write a Java program to display the name and path of a file.
4. Write a Java program to write data into a file using FileWriter.
5. Write a Java program to read data from a file using FileReader.